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Internal business process governance and external regulation: How does AI technology empower financial performance?☆

Xuanmei Cheng [a], Anna Min Du [b], Chengnuo Yan [a], John W. Goodell [c,*]

[a] School of Management, Zhijiang College, Zhejiang University of Technology, Hangzhou, China
[b] The Business School, Edinburgh Napier University, Edinburgh, UK
[c] College of Business, The University of Akron, Akron, OH, USA

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ABSTRACT

We examine the impact of artificial intelligence (AI) technology on the financial performance of financial firms, focusing on the transmission role of business process governance from the perspective of corporate governance. We examine whether there is heterogeneity in the effects of different institutional environments, conducting an empirical analysis of the sample data of 152 Chinese financial firms from 2011 to 2022. AI technology improves financial enterprises' operational quality through risk management, internal governance, and internal control. Moreover, AI provides technical support to enhance service derivative capability by enhancing the number of businesses across cross-regional operations, resource competition, and customer excavation. We also identify that the enabling effect of digital technology on business process governance is affected by imitation pressure from peer firms. In contrast, the impact of government regulatory pressure is not significant.

1. Introduction

We focus on the role of artificial intelligence (AI) and digital technology in facilitating firm business process governance, highlighting the aiding of business process governance as a major aspect of AI firm integrations. The important role of business process governance has been emphasized in recent literature, including Guo et al. (2024) (taxation management), Lu et al. (2023) (supply chain management), Gao et al. (2022) (outward foreign direct investment), and Su et al. (2022) (food safety and production governance).

Artificial intelligence has become ubiquitous. From intelligent voice assistants to self-driving cars, smart home systems, and medical diagnostics, the application of artificial intelligence (AI) continues to change how we produce, live, and learn. Domestic industries have similarly recognized the significance of AI technology in leading a new wave of scientific and technological revolution and industrial change and have adjusted their development strategies to conduct innovative activities using AI technology (Liu et al., 2020; Zhai & Liu, 2023). In the financial sector, AI has demonstrated significant technological value in obtaining incremental business, improving operating costs, optimizing data resource management, and boosting the availability and effectiveness of financial services in product design, information security, risk control, and customer service (Alaimo & Kallinikos, 2021). Globally, the financial AI industry market continues to grow, showing immense potential for multiple segments. According to the IMARC Group, the global financial AI market reached $14.6 billion in 2023 and is expected to grow to $87.7 billion by 2032, growing at a CAGR (Compound Annual Growth Rate) of 20.9 %. For this reason, the Chinese government attaches great importance to developing this technology and has put forward the strategic plan of "taking a new generation of artificial intelligence as a driving force to promote the development of science and technology across the board, the optimization and upgrading of industries, and the overall leap in productivity, and striving to achieve high-quality development." They aim to vigorously promote the deep integration of AI in the real economy, giving full play to the great potential of artificial intelligence technology in promoting the leapfrog development of science and technology, the optimization and upgrading of industry, and an overall leap in productivity (Wang & Li, 2024). Recent studies agree that the transformation of the financial landscape through AI has significant economic value (Goodell et al., 2021; Younis et al., 2024). For example, the emergence of new financial service models such as digital inclusive finance, fintech (Broby, 2021), and so

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* Corresponding author.
E-mail addresses: may1981@zjut.edu.cn (X. Cheng), a.du@napier.ac.uk (A.M. Du), johngoo@uakron.edu (J.W. Goodell).

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forth, helps financial institutions enhance their risk-bearing capacity (Tang et al., 2024), reduce financing costs, and improve their performance. Therefore, exploring AI-technology-enabling mechanisms is crucial for achieving high-quality development in the financial industry.

A McKinsey & Company report pointed out that in recent years, the global scope of the enterprise AI utilization rate has increased significantly. An average of 50 % of the companies deployed AI in 2022, much higher than the 20 % deployed in 2017. However, China's current AI utilization rate is only 41 %, temporarily lagging behind the global average; thus, there is room for improvement. AI technology is still in the exploratory stage in the financial industry, a data-intensive industry with a strong demand for digitalization. Moreover, an imbalance exists in intelligence development among enterprises of varying sizes. There is also a great step in this technology's large-scale application and realizing its commercial value. This is because various scenarios in the financial field must be calculated with high precision to arrive at a rigorous answer with low tolerance for error (Inklaar & Yang, 2012). Algorithmic errors or data bias may lead to inaccurate investment advice. Therefore, the potential algorithmic and data security issues associated with AI technology require financial enterprises to have strong data operation capabilities. Moreover, as the incremental business volume of the financial industry gradually shrinks, refined operations have become the current goal, and the original business operation mode is no longer applicable (Chen et al., 2021; Lee et al., 2021). Based on these viewpoints, financial firms urgently need to leverage AI technology as a powerful tool to strengthen their internal business process governance and achieve the operational goals of reducing costs and increasing efficiency.

This study examines the impact of technology-driven business process governance on performance from a corporate governance perspective. It highlights how AI can enhance financial performance in the industry and alleviate the economic pressures stemming from technological applications. The findings provide theoretical and practical insights for improving performance and leveraging digital technology's spillover effects in governance. However, financial firms face social pressure from the external environment (Amer, 2022), meaning that institutional pressures influence AI applications and business decisions. Specifically, financial firms must comply with industry regulations when using AI for service improvement and process optimization. Economic uncertainty, privacy risks, algorithm instability, and data security issues may hinder organizational change, underscoring the need for regulation. Firms must also monitor peers to maintain a competitive edge. Based on institutional theory, this study explores how external pressures from government and market forces moderate the relationship between AI and business process governance, enriching the understanding of the conditions affecting governance decisions.

This study explores the internal mechanisms of the impact of AI technology on firms' financial performance using panel data on financial firms listed as A-share companies in China from 2011 to 2022. AI technology can enhance firms' digital operational and service derivative capabilities, thus improving financial performance. Specifically, the application of AI technology in the financial industry enhances the quality of existing businesses by improving the performance of enterprises in cost control, digital governance, and risk avoidance. It also provides technical support for enterprises to break geographical constraints, compete for financial resources across regions, and expand customer relationships, for which enterprises are willing to take the initiative and risk of developing new businesses. Additionally, we find that the impact of AI technology on corporate governance capabilities varies under different degrees of industry competition.

The theoretical contributions of this study are as follows. First, it reveals AI's role as an innovation driver in promoting financial performance and its underlying mechanisms. While previous research in the financial sector focused on factors like policy guidance (e.g., green credit) or external evaluations (e.g., CSR, ESG), the potential economic benefits of AI as a performance enhancer remain underexplored. This study addresses the unclear mechanisms behind technology-enabled performance improvements. Second, current research on performance enhancement often centers on innovation and strategy, which do not fully explain AI's impact on financial business governance. This study explores how digital technology improves performance through enhanced business process governance. While some studies have examined digital innovation's role in risk-taking and cost control, evaluations of its effects are mixed. To clarify this, the study subdivides business process governance into business quality (improving business operations) and business quantity (expanding operations), examining the mediating role this plays between AI and financial performance. Finally, the study analyses how regulatory pressure from the government and imitation pressure from peers influence AI's impact, expanding the application of institutional theory. While most research has focused on manufacturing and high-tech industries, this study highlights that financial firms also face external "social pressure" affecting their performance.

The development of AI has brought opportunities and challenges to the financial industry. AI technology empowers the development of the financial industry, realizes comprehensive coverage of major financial scenarios, and continues to promote intelligent transformation, thus playing a key role in promoting the development of the entire industry. However, in practical applications, the compatibility of AI technology with the current financial operation model needs to be improved further, and financial enterprises need to conduct business governance to cope with technological changes actively. Based on the results of this paper, financial enterprises should take the initiative to explore and fully release the practical value of AI technology to reshape the new financial business format comprehensively. Relevant financial regulatory authorities should also actively implement incentive measures to create an environment conducive to innovation and promote industry development.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature and develops the research hypotheses. Section 3 outlines the research methodology, data sources, and variable definitions. Section 4 presents the empirical results. Section 5 discusses endogeneity and robustness tests. Finally, Section 6 presents conclusions and practical implications.

2. Theoretical analysis and research hypotheses

2.1. The impact of AI technology on financial performance

With the rapid advancement of AI, its deep integration with the financial sector has transformed traditional financial operations and had disruptive effects on the entire market. How will digital technology reshape the financial industry and contribute to its high-quality development in practical scenarios? Existing research highlights the importance of technological upgrades in finance, focusing on the impact of tech-driven innovation on corporate social responsibility, business performance, and more. For instance, Cheng et al. (2023) show fintech's positive influence on the real economy by reducing carbon emissions, while Lin et al. (2023) argue that digital finance improves corporate investment efficiency. However, the role of technology as a key driver of innovation remains unclear. Research on financial performance drivers has primarily focused on policy initiatives like green credit and external evaluations like ESG, often overlooking the economic benefits of technological innovation. This study aims to address this gap by exploring the impact of AI on financial performance, clarifying its application in financial scenarios, and providing theoretical guidance for the sector's rapid development and value creation.

With artificial intelligence, cloud computing, and other emerging technologies, financial enterprises are undergoing a comprehensive and in-depth digital transformation from internal businesses to external operations. Based on the intelligent transformation of the financial industry by AI technology and its application in actual scenarios, this

study argues that financial institutions can use digital technology to empower scenario innovation and provide technical support for data element activation and maturity matching, thus contributing to an improvement in their financial performance through service optimization, value release, and risk avoidance.

First, AI application scenarios cover all aspects of the financial industry chain, including product design, investment advisory services, and customer relationship management. Optimizing the service model promotes the online intellectualization of financial products and services, breaks the limitations of geographic distance and service costs, and improves the accessibility of financial services. Moreover, AI technology is an effective tool for opening the connection channel between financial institutions and the market. This allows funds to flow into a solid economy by integrating multiple parties to build an industrial ecosystem, realize the resonance of mode and technology, and promote the optimization and upgrading of the industrial structure and high-quality development of the financial industry. In addition, effective information mining of massive amounts of data through digital technology provides data support for financial enterprises to conduct related businesses, such as forward-looking prediction of market trends and layout in advance, accurate identification of the individual needs of customers, and providing them with customized financial products and services to enhance the financial performance of enterprises.

Additionally, AI can improve the management and operation of financial enterprises' data assets, extract effective business information from unstructured daily operations data, strengthen data analysis capabilities, and expand the scope of data sharing. The link between assets is visualized, providing a reliable basis for investment and financing inside and outside the industry. Secondly, AI technology integrates and centralizes relatively dispersed data information of enterprises and filters out redundant information after comprehensive computing to improve the quality of data information and build asset images, solving the problems of searchable and comprehensible data assets and releasing the value of data fully. Further, using AI technology for data capture to obtain visualized and integrated information can form a scale effect (Sjödin et al., 2021). Relying on the data assets allows high-level financial product development, related investment and financing risk prediction, and other benefits, but also achieves rapid data to the demand side of the reach. The service network extends to industries, which helps improve the financial performance of financial enterprises on the supply side.

Further, applying digital technology and AI can enhance data timeliness and address inefficient processes. By comparing and analyzing business data, such as corporate financial statements and tax information (Nayal et al., 2021), real-time monitoring of business activities can be achieved, enabling timely dynamic adjustments. Second, AI technology can mitigate information asymmetry between supply and demand by using intelligent matching, optimizing the credit term structure, and improving access to long-term credit (Chen, Chu, & Zhao, 2023). This helps businesses better align long-term investments with their financing, reducing the risks associated with mismatches between investment horizons and liabilities. Additionally, AI algorithms can predict potential credit violations in advance, assisting financial enterprises in formulating more effective risk control strategies. By implementing intelligent risk monitoring systems, businesses can conduct pre-credit investigations, credit reviews, and post-credit inspections, gaining comprehensive insights into risk information. This, in turn, provides a secure technological framework to enhance financial performance and offer valuable insights into risky business activities and platform ecosystems. Based on these views, this study proposes the following hypothesis:

H1. *Financial firms' AI level is positively related to financial performance.*

2.2. Mediating effects of business process governance

The application of AI technology in the financial industry has significant potential, such as a gradual rise in spatial finance, one-stop financial service platforms, and computationally intensive transaction services, which have become important engines for industrial development. Scholars believe that the entry of AI is an opportunity, yet it also carries considerable risk (Broekhuizen et al., 2023). Enterprises are driven by economic pressure to introduce new digital technologies, and the use of these technologies will impact their business management and service models, thus affecting their performance (Chen & Xu, 2022), mainly from the perspectives of innovation and strategy. Wan et al. (2023) discussed the enabling effect of digital technology on innovation performance and the transmission mechanism of R&D investment and technology diffusion. Zheng et al. (2024) argued strategically that digital business strategies can improve performance by enhancing dynamic knowledge capabilities. In addition, studies have shown that apart from "economic pressure", there are spillover effects of the technology itself that contribute to changes in behavior. Scholars have explored the relationship between the innovative application of digital technology and the role of financial institutions' internal governance, such as risk-taking and cost control. For example, Wu et al. (2023) compared the impact of fintech on several types of risk, finding that fintech can help enterprises effectively control credit risk and reduce bankruptcy risk. Scholars believe that emerging technologies exacerbate cybersecurity risk, and enterprises often increase cyber technology expenditure based on strategic needs, which harms stability (Uddin et al., 2020). However, no consensus has been reached regarding the effects of digital technology.

Based on the theoretical perspective of corporate governance, this study selects digital operational capability and service derivative capability to characterize the business process governance of financial firms from both qualitative and quantitative dimensions. We test the role of business process governance in the impact of AI technology on financial performance and analyze the driving mechanism of digital technology on improving their performance. Digital operational capability is reflected in enhancing risk management efficiency, improving internal governance and control mechanisms, constructing business knowledge management systems, and other aspects of business quality (Su & Wu, 2023). Additionally, the use of AI technology by financial enterprises to actively conduct cross-regional operations, resource competition, customer mining, and other related business expansion behaviors can empower financial enterprises to improve their credit levels, which, in turn, leads to the investment of more credit resources. Enterprises' service derivative capability supports this process.

2.2.1. Digital operational capability

Financial enterprises use AI technology to improve their digital operational capability and achieve financial performance breakthroughs using business quality improvement, mainly in the following aspects. From the cost control perspective, AI, with its efficient, personalized, and low-cost features, quickly integrates and updates customer-related information and adopts data-mining algorithms to assess customer risks before pushing and promoting products accurately. This can reduce financial institutions' credit approval and marketing costs (Kumar et al., 2024). By constructing an intelligent investment advisory platform, AI can replace middle- and low-end analysis on a large scale, make predictions in the time dimension, reduce the emotional interference of human beings, and optimize the decision-making effect and performance enhancement. From the perspective of digital governance, based on big data processing and analysis technologies, it performs regular operation and governance of data assets to achieve "digital governance with digital technology". Based on operational indicators, relevant technical personnel can monitor data quality while carrying out stage-by-stage task deployment and tracking feedback, driving financial business digital transformation and standardized operations and helping

improve corporate performance. From the perspective of risk avoidance, financial credit risk is primarily caused by information asymmetry due to internal agency conflicts (Derrien et al., 2016), The introduction of AI technology can open up an information flow path between various levels to alleviate this problem. In addition, compared to artificial and intelligent risk control, other technologies can carry out round-the-clock dynamic supervision, facilitating customer data security, real-time monitoring of business operation status, and identifying systematic risks. Moreover, with the powerful deep learning and iterative evolution capabilities of AI tools, reliable risk control models can be constructed to identify potential risks from pre-to post-lending accurately and then formulate effective risk mitigation measures to achieve full-cycle risk avoidance and performance growth (Habbal et al., 2024). We introduce the following hypothesis:

H2a. *The level of AI technology in financial firms improves financial performance by enhancing digital operational capability*.

2.2.2. Service derivative capability

Financial enterprises rely on AI technology to enhance their service derivative capability and improve financial performance by expanding their business scope in the following ways. From the perspective of risk-taking, previous studies conclude that it is difficult for financial institutions to maintain the stability of information systems, personnel structure, and other organizational elements when expanding (Ballouk et al., 2024), and the difficulty of operation is significantly increased, which is not conducive to corporate management and control, and that enterprises will reduce such behaviors based on risk considerations. However, the deep integration of AI technology and financial business can provide technical guarantees for enterprises to carry out cross-regional operations, cross-border cooperation, and other business expansion behaviors and enhance risk-bearing initiatives (Li et al., 2022). Business diversification increases the likelihood of achieving performance targets. From the resource allocation perspective, financial enterprises must invest substantial resources in exploring new businesses. The AI middle platform they built can provide powerful AI capabilities and algorithmic model support for the aggregation and configuration of the resources of various elements and the construction of business scenarios to reduce the waste of resources and enable enterprises to invest in innovation to remain stable. The application of AI technology is also conducive to the acquisition, flow, and sharing of data resources, thereby improving the efficiency of resource reserves and utilization (He et al., 2024; Li et al., 2023), which provides a basis for clarifying the direction of business development. Correct business decision-making is conducive to the growth of financial performance. From the perspective of customer relationships, financial enterprises have embedded new applications of artificial intelligence on corresponding information technology, launched an emotional model session business to enhance customer insight capability, strengthened customer retention capability on the business side through deep data mining and intelligent reach, and enhanced the ability to stick to customers with personalized AI voice synthesis and automated generation of marketing creative copies. The extension and stabilization of customer networks have been realized, providing a customer base for performance improvement (Frank et al., 2021). Based on these views, this study proposes the following hypothesis:

H2b. *The level of AI technology in financial firms improves their financial performance by enhancing their service derivative capability.*

2.3. Moderating effects of institutional pressure

According to institutional theory, financial firms are embedded in the institutional environment, and their behavioral decisions are not based solely on the rational consideration of costs and benefits. Still, they are also influenced by the behavior of other financial institutions and the demands of stakeholders in the environment, i.e., the "social pressure" exerted by the external environment. Under this pressure, financial firms' business process governance must comply with legitimacy requirements rather than pursue profit maximization to achieve social resilience. Simultaneously, organizational legitimacy, as a key external resource for social returns, can be a powerful catalyst for other activities. Existing research applies institutional theory to the study of corporate environmental responsibility, focusing on manufacturing enterprises, high-tech enterprises, and other groups, exploring how different dimensions of institutional pressure affect organizational behavior and how the institutional environment moderates the relationship between the characteristics and capabilities of enterprises and their performance. For example, Siyal et al. (2023) explore the internal mechanism of institutional pressure to promote manufacturing enterprises' green development and transformation from the perspective of legitimacy. Zeng et al. (2017) explore the mechanism and relationship between institutional pressure, supply chain relationship management, sustainable supply chain design and circular economy capability of Chinese eco-industrial park enterprises and concluded that different pressures would have different degrees of negative regulation on enterprises. Huang and Chen (2022) explore the mechanism and relationship among institutional pressure, supply chain relationship management, sustainable supply chain design, and circular economy capability of Chinese eco-industrial park enterprises. High-tech enterprises explore the mechanisms underlying institutional pressure and slack in corporate green product innovation. This study focuses on the financial industry to expand its application to institutional theory.

As the behavior of enterprises in China comes more from the response to the government and the market, the institutional pressure discussed in this study is regulatory pressure from the government and imitation pressure from peer enterprises and competitors in the market (Du et al., 2023). Specifically, the organizational structure and decision-making mode of financial enterprises need to comply with current social expectations and legal policies and also accept the mandatory control of government departments (Zhu et al., 2023). In the face of environmental uncertainty, financial enterprises will selectively imitate the behavior of other similar organizations or benchmark organizations, which is an incentive for the market to guide. However, enterprises are subject to the same pressure to catch up. Based on the above viewpoints, this study constructs a theoretical framework of AI technology for business process governance improvement under different institutional pressure scenarios to enrich the research on digital technology applications and business development management.

2.3.1. Regulatory pressure

Financial regulation refers to the government's restrictions or regulations on actors in financial transactions through specific institutions. The implementation of financial supervision by the government can regulate the business behavior of financial enterprises in the process of intelligent transformation, especially during new technology entry. First, financial supervision controls the risky behaviors brought about by technology; specifically, it regulates service quality, risk management, information security, and other aspects of financial enterprises and requires them to ensure that the data are legally obtained and correctly used. Guiding financial firms' behavioral compliance and technology risk control can stimulate them to optimize and upgrade business. Zhang et al. (2023) confirm that government regulations can help improve the governance of firms' financial investments and financialization. Second, financial regulation will strengthen the "targeting" of financial services. Financial regulations and policies often lag behind the rapid development of finance, especially in the face of the rapid iteration of AI technology, which is likely to threaten the stability and resistance of the economic system. Therefore, strengthening financial regulations can help ensure effective regulation and timely correction of corporate governance. Zheng et al. (2023) suggest that relevant digital financial regulatory policies improve investment efficiency by reducing inefficient investments and underinvestment in financial enterprises,

reflecting the strengthening of corporate governance capacity. Thus, we propose the following hypothesis:

H3a. *Regulatory pressure can strengthen the positive effects of AI technology on digital operational capability.*

2.3.2. Imitation pressure

Imitation pressure, as a weak legitimacy mechanism, affects organizational behavior by influencing resource allocation or incentives (Martínez-Ferrero & García-Sánchez, 2017). Financial enterprises typically adopt the most prudent strategy to avoid lagging when facing high market uncertainty and complex AI-related technologies. First, as an external governance mechanism, market competition exerts pressure on internal governance and continuously urges enterprises to improve their internal control mechanisms. Scholars have also confirmed that external threats, such as competition, can put enormous pressure on enterprises, prompting managers to implement business management to improve performance. For example, in the face of departmental barriers and information silos within an enterprise, real-time data warehousing can be constructed to improve the timeliness and usefulness of data resources. Given the vacuum in internal management caused by the development of new products and businesses, financial institutions will continue to improve their internal control and governance systems, effectively control potential risks and problems, and deepen the integration of AI with customer services, product promotion, and other scenarios. Second, scholars proposed that imitation pressure will positively regulate the relationship between technology and firm innovation behaviors (Huang et al., 2022). Lin and Ho (2011) argued that business executives often interpret imitation pressure as a business opportunity brought about by market demand. Enterprises respond more positively than they do under regulatory pressure, show a stronger willingness to change, and take the initiative to explore new business and financial products, such as accelerating the construction of new technology platforms and exploring new business scenarios such as smart branch operations, internal knowledge services, and auxiliary coding. For example, they accelerated the construction of new technology platforms and attempted to explore scenarios such as intelligent security, branch operations, internal knowledge services, and assisted coding, which resulted in continuous breakthroughs in the coverage of business scenarios. Based on these views, this study proposes the following hypothesis:

H3b. *Imitation pressure can strengthen the positive effects of AI on service derivative capability.*

3. Research design

3.1. Sample selection and data acquisition

We consider financial enterprises among Chinese A-share listed companies in Shanghai and Shenzhen as research objects, including banks, securities companies, and insurance companies, with a sample period of 2011–2022. This research interval is chosen because "*the White Paper on the Transformation of the Banking Industry in the Era of Intelligent Finance*" pointed out that since 2011, the improvement in computing power and the exponential growth of data volume have led to the explosive development of artificial intelligence. By the end of 2022, the emergence of ChatGPT opened a new era of artificial intelligence; therefore, 2023 is a dividing point and cannot be classified as a stage with previous years.

The research samples are screened according to several steps. First, we obtain a list of Shanghai and Shenzhen A-share listed companies through the Cathay Pacific Listed Companies Database (*CSMAR*), and then screen all companies in the financial sector according to industry classification, We then integrate the required financial data. The annual reports of the listed companies are collected and collated from the *Sina Finance* website. We use machine learning methods to perform textual analysis and word frequency statistics to generate AI dictionaries and construct enterprise AI indicators. In addition, financial performance data is from *CSMAR*, and measurement of digital operational capability from Wind. Measurement of service derivative capacity is from *CNRDS*; measurement of institutional pressure from the National Bureau of Statistics, and control variables from *CSMAR* and *CSDN*. To ensure the validity of the data, samples with missing data, ST companies, and only one year are excluded, shrinking the tail by 1 %. Finally, we obtain unbalanced panel data containing 1114 observations from 152 financial companies over 12 years.

3.2. Variable selection

3.2.1. Dependent variable: corporate financial performance

Currently, the academic community usually uses the Tobin Q value and return on total assets (ROA) to measure the financial performance of enterprises. The Tobin Q value can better reflect the market performance of enterprises. Therefore, this study selects the Tobin Q value in the initial regression model to measure the financial performance of enterprises.

3.2.2. Independent variable: AI technology

With regard to the construction of AI technology indicators, this study refers to the practice of Chu et al. (2024), who disassemble the text of the annual reports of listed companies collected from the Sina Finance website; the patent text collected from *IRPDB* intellectual property database for content disassembly and word frequency statistics generates 73 AI-related words by eliminating repetitive and low-frequency words and then constructs corporate AI indicators by adding the number of AI keywords in the text of the annual report by 1 to take the natural logarithm. An enterprise artificial intelligence index is constructed. To ensure the robustness of the results, we construct a proxy indicator for corporate AI based on the "Management Discussion and Analysis" (*MD&A*) section of the annual report. We also construct the proxy indicator by adding 1 to the number of AI keywords in the *MD&A* section and taking the natural logarithm of the number of AI keywords in the *MD&A* section.

3.2.3. Mediator: digital operational capability and derivative capability

Digital operational capability refers to the ability of enterprises to operate efficiently and respond flexibly by leveraging digital technologies. Warner and Wäger (2019) design scale items in their research, including "companies can use digital means to optimize business processes or resource allocation," "companies can carry out real-time dynamic analysis of services and resources for flexible adjustment," "companies have improved the efficiency of business intelligence decision-making through digital tools and components," etc., indicating that enterprises with strong digital operational capability can optimize resource allocation and product structure, drive decision-making with data, and improve the overall business quality. In this process, the speed of capital turnover and the utilization effect were significantly improved. Moreover, financial companies' businesses are digitized. Therefore, this study selects the total asset turnover rate, which can comprehensively evaluate enterprises' operational quality and asset utilization efficiency at the financial level, as a proxy for digital operational capability.

For financial enterprises, derivative capability is an extension of the concept of service-oriented manufacturing (Liu et al., 2023). If an enterprise wants to develop a new business, it must fully exploit the advantages of data, technology, and other factors. We use the natural logarithm of the total number of invention patents, utility models, and design patents independently applied for by enterprises in the current year (the natural logarithm is taken after adding 1 to the number of patent applications) as the measure of this variable. Of course, what constitutes derived capabilities can be far more varied. However, we feel this control variable is reasonably parsimonious to consider such a broad sweeping concept.

3.2.4. Moderator: regulatory pressure and imitation pressure

The regulatory pressure on financial enterprises comes from the government's supervision and control. Therefore, to truly reflect the strictness of the government's regulation of the financial industry, this study focuses on the input of financial affairs in fiscal expenditure. Based on the principle of territorial supervision of small- and medium-sized financial institutions and the availability of data, this study adopts the ratio of regional financial supervision expenditure to the financial industry's value added as a benchmark indicator of financial supervision intensity, referring to the practice of Wang and Hu (2023). The data are then matched according to the province to which the financial enterprises belong.

The imitation pressure on financial firms comes from other firms in the industry and embodies a kind of competitive market pressure. Based on previous studies, we use the Herfindahl-Hirschman index (*HHI*) to measure the degree of competition in an industry (Abdoh & Varela, 2017). *HHI* is usually defined as the sum of the squares of firms' market share in an industry. The formula is $HHI = \sum [(Xi / X)^2]$, which is calculated by using the main business revenue of individual firms to calculate their market share in the industry. Here, X_i is the enterprise's main business revenue and X is the industry's main business revenue. The smaller the *HHI*, the more intense the market competition in the industry and the greater the imitation pressure perceived by management. Therefore, this study adopts the opposite of the *HHI* index to characterize imitation pressure on enterprises.

3.2.5. Control variables

Referring to related studies, because many factors affect the financial performance of enterprises, the control variables selected in this study include enterprise size (*Size*), age of the enterprise (*Age*), financial leverage (*Lev*), nature of property rights (*Soe*), growth of the enterprise (*Growth*), shareholding concentration (*Sh*), concurrent two-ranking (*Dual*), and size of the board of directors (*Board*). In addition, the effects of individuals and time (*Year*) are controlled. Table 1 presents these variables.

3.3. Econometric model design

3.3.1. Benchmark modeling

To test the hypothesis on the impact of AI technology on firms' financial performance, we employ a two-way fixed effects model that controls for the impact of individual differences and temporal trends on outcomes. This enables an effective understanding and analysis of the relationships and interactions between the variables in the research model and reveals the internal logic of the economic structure. To show the relationship between the variables more intuitively, we construct an econometric model expressed as follows:

$$CFP_{i,t} = \beta_0 + \beta_1 AI_{i,t} + \beta_2 Controls_{i,t} + \lambda + Year_i + \varepsilon_{i,t} \quad (1)$$

Here, $CFP_{i.t}$ is the explanatory variable representing the financial performance of firm i in year t; $AI_{i.t}$ is the core explanatory variable representing the level of AI technology of firm i in year t; Controls are the other control variables, λ_i denotes the individual fixed effect, $Year_i$ denotes the time fixed effect, and $\varepsilon_{i,t}$ is the error disturbance term.

3.3.2. Mediator modeling

We use digital operational and service derivative capabilities as mediating variables and establish the model as follows:

$$DOC_{i,t} = \gamma_0 + \gamma_1 AI_{i,t} + \gamma_2 Controls_{i,t} + \lambda_i + Year_i + \varepsilon_{i,t} \quad (2)$$

$$CFP_{i,t} = \beta_0 + \beta_1 DOC_{i,t} + \beta_2 AI_{i,t} + \beta_3 Controls_{i,t} + \lambda_i + Year_i + \varepsilon_{i,t} \quad (3)$$

where *DOC* denotes the firm's digital operational capability, and the meanings of the other variables are consistent with those in the baseline model. Model (2) tests the impact of AI technology on firms' digital operational capability. Model (3) tests the impact of firms' AI technology levels on their financial performance and controls for digital operational capability:

Table 1
Variable definitions.

Variable	Symbol	Definition
Independent variables		
Artificial intelligence technology	*AI*	Analyze the self-organized text materials and make word-frequency statistics. Use the number of keywords plus 1 to take the logarithm.
Dependent variables		
Corporate financial performance	*CFP*	Tobin Q = market capitalization/ total assets
Mediators		
Digital operational capability	*DOC*	Total asset turnover
Service derivative capability	*SDC*	The total number of invention patents, utility models, and design patents independently applied for by the enterprise in the current year is added by 1 to take the natural logarithm
Moderators		
Regulatory pressure	*RP*	The ratio of the scale of regional financial regulatory expenditure to the value-added of the financial sector.
Imitative pressure	*IP*	The inverse of the *HHI* index
Control Variables		
Firm size	*Size*	The logarithm of the total assets
Firm age	*Age*	The length of the number of years from the date the enterprise was established to the statistical year is taken as the natural logarithm.
Leverage	*Lev*	Measured by the debt-to-asset ratio, it is calculated by dividing total liabilities by total assets.
Growth	*Growth*	Measured by the growth rate of total operating income, it is calculated by dividing the growth of operating income by the total operating income of the previous year.
Board size	*Board*	The number of board members is taken as the natural logarithm
Duality	*Dual*	If the chairman of the board of directors and the general manager of the enterprise are the same person, the value is assigned as 1. Otherwise, the value is 0.
Ownership concentration	*Sh*	The shareholding ratio of the largest shareholder
Property	*Soe*	Whether the enterprise is a state-owned enterprise or not, if it is a state-owned enterprise, the value is 1, otherwise it is 0
Year fixed effect	*Year*	Dummy variable

This table shows variable definitions and sources.

$$SDC_{i,t} = \gamma_0 + \gamma_1 AI_{i,t} + \gamma_2 Controls_{i,t} + \lambda_i + Year_i + \varepsilon_{i,t} \quad (4)$$

$$CFP_{i,t} = \beta_0 + \beta_1 SDC_{i,t} + \beta_2 AI_{i,t} + \beta_3 Controls_{i,t} + \lambda_i + Year_i + \varepsilon_{i,t} \quad (5)$$

where SDC denotes the firm's service derivative capability, and the meanings of the other variables are consistent with those in the benchmark model. Model (4) tests the effect of AI technology on firms' service derivative capability. Model (5) tests the effect of firms' AI technology levels on financial performance by controlling for service derivative capability.

3.3.3. Moderator modeling

In this study, we use regulation and imitation pressures as moderating variables and introduce cross-multiplier terms into the following model:

$$DOC_{i,t} = \beta_0 + \beta_1 AI_{i,t} + \beta_2 AI_{i,t} \times RP_{i,t} + \beta_3 Controls_{i,t} + \lambda_i + Year_i + \varepsilon_{i,t} \quad (6)$$

$$SDC_{i,t} = \beta_0 + \beta_1 AI_{i,t} + \beta_2 AI_{i,t} \times IP_{i,t} + \beta_3 Controls_{i,t} + \lambda_i + Year_i + \varepsilon_{i,t} \quad (7)$$

where *RP* denotes regulatory pressure, *IP* denotes imitation pressure,

and the other variables are the same as those in the benchmark model. Model (6) tests the moderating effect of regulatory pressure on the relationship between AI technology and digital operational capability, and Model (7) tests the moderating effect of imitation pressure on the relationship between AI technology and service derivative capability.

4. Empirical analyses

4.1. Descriptive statistics

Table 2 presents the descriptive statistics of the variables in the regressions. The maximum and minimum values of *CFP* are 8.1629 and 0.8452, respectively, indicating differences in financial performance among different financial firms and that firms are operationally inefficient and cannot make ends meet. Meanwhile, the value range of the independent variable *AI* is 0.0000–3.6109, and the standard deviation is 1.0267, indicating a gap in the level of AI technology among different enterprises. The mediating variables *DOC* and *SDC* take values of 0.0105–0.7599 and 0.0000–6.6053, respectively, indicating significant differences in business process governance, especially the digital operation capability, of different enterprises and the same enterprise in different years. The values of *RP* and *IP*, on the other hand, indicate that enterprises are subject to different degrees of governmental regulation and industry competition because of the regions in which they are located.

4.2. Analysis of empirical results: benchmark regression and mediation effects

Table 3 reports the results of the impact of AI technology on financial firms' financial performance under the two-way fixed effects panel model. The results of Model 2 show that the regression coefficient of AI technology (*AI*) on financial firms' financial performance (*CFP*) is significantly positive ($\beta = 0.1016$, $p < 0.01$), indicating that financial firms can effectively contribute to the improvement of their financial performance through the introduction of AI technology, and Hypothesis 1 is verified. AI technology has a profound impact on the development of the financial industry. The financial industry should actively embrace changes, have a positive learning attitude towards AI technology, and continue to promote the application of AI technology in financial services. This will enhance its core competitiveness and promote the development of the financial industry in a more intelligent, efficient, and personalized direction.

The results of Model 3 show that the regression coefficient of AI technology (*AI*) to the digital operational capability (*DOC*) of enterprises is significantly positive ($\beta = 0.0087$, $p < 0.05$), indicating that the integration of AI technology and financial scenarios is conducive to enhancing the digital operational capability of financial enterprises. This manifests in improving enterprise risk management level, internal governance, internal control efficiency, etc.

After further incorporating the intermediary variable *DOC*, Model 4 shows that the regression coefficient of AI technology (*AI*) is still significantly positive ($\beta = 0.0896$, $p < 0.05$), but the impact coefficient is lower than that of Model 2. This indicates that enterprises' digital operational capability plays an incomplete mediating role between AI technology and financial performance, thus verifying Hypothesis 2a.

This is because AI technology empowers the digital operational capability of enterprises, with information barriers opened and decision-making errors and internal conflicts caused by information asymmetry mitigated to achieve risk aversion. Technology replaces low-end work and shrinks the operating costs of enterprises. Through the digital intelligent transformation of financial business, the quality of products enterprises provide and the success of investment and financing are significantly improved. Overall, business quality was optimized and improved through the rational use of digital technology, resulting in improved financial performance.

In addition, the results of Model 5 show that the regression coefficient of AI technology (*AI*) to enterprise service derivative capability (*SDC*) is significantly positive ($\beta = 0.1620$, $p < 0.01$), indicating that AI technology can enhance the service derivative capability of financial enterprises, which is manifested in the application of large models to significantly enhance the ability of enterprise customer mining, resource acquisition, and business expansion.

The results of Model 6 show the regression results after the inclusion of the mediating variable *SDC*, and the results show that the regression coefficient of AI technology (*AI*) is still significantly positive ($\beta = 0.0919$, $p < 0.05$). However, the impact coefficient is reduced compared to that of Model 2. This indicates that enterprises' service derivative capability plays an incomplete mediating role between AI technology and financial performance, validating Hypothesis 2b.

The specific reason may be that financial enterprises carry out business governance with technological support and are willing to take the initiative to take risks, with the motivation to pursue business performance. Enterprises use algorithms to optimize resource allocation, and the flow of internal and external resources is accelerated. In addition, AI technology enables intelligent identification of customers, and with the expansion of the financial services group, enterprises can actively explore new businesses to meet the individual needs of customer personalization, and performance can be significantly improved. The potential reason for the partial mediation effect is that financial enterprises improve their AI level, which may not only empower financial performance by enhancing digital operational and service derivative capabilities but also promote improving financial performance by strengthening risk management and cost control

Table 2
Descriptive statistics.

Variables	N	Mean	SD	P25	P50	P75	Min	Max
AI	1114	1.0197	1.0267	0.0000	0.6931	1.7918	0.0000	3.6109
CFP	1114	1.5126	1.1574	0.9928	1.0977	1.4867	0.8452	8.1629
DOC	1114	0.1232	0.1624	0.0267	0.0553	0.1406	0.0105	0.7599
SDC	1114	0.6236	1.3540	0.0000	0.0000	0.6931	0.0000	6.6053
RP	1114	0.0070	0.0076	0.0025	0.0044	0.0080	0.0009	0.0386
IP	1114	−0.1793	0.1492	−0.1869	−0.1116	−0.0958	−0.7892	−0.0682
Size	1114	25.4922	2.6541	23.6037	25.3738	27.3366	19.9203	30.9344
Lev	1114	0.7147	0.2204	0.6067	0.7639	0.9152	0.0809	0.9473
Growth	1114	0.1764	0.5907	−0.0597	0.0794	0.2575	−0.7219	3.8116
Board	1114	2.3572	0.3007	2.1972	2.3979	2.5649	1.6094	2.8904
Dual	1114	0.1481	0.3554	0.0000	0.0000	0.0000	0.0000	1.0000
Sh	1114	0.2951	0.1626	0.1827	0.2515	0.3533	0.0628	0.7230
Soe	1114	0.6176	0.4862	0.0000	1.0000	1.0000	0.0000	1.0000
Age	1114	3.1097	0.2921	2.9957	3.1781	3.2958	2.0794	3.5835

This table shows the number of observations, the means, standard deviations, and 25, 50, and 75 percentiles, as well as the minimums and maximums of our study variables. Variable definitions are in Table 1.

Table 3
Basic and mediating effects.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	CFP	CFP	DOC	CFP	SDC	CFP
AI		0.1016***	0.0087**	0.0896**	0.1620***	0.0919**
		(0.0388)	(0.0039)	(0.0366)	(0.0483)	(0.0378)
DOC				1.3774***		
				(0.5280)		
SDC						0.0601***
						(0.0168)
Size	−0.7555***	−0.7466***	−0.0264	−0.7103***	−0.0089	−0.7461***
	(0.1880)	(0.1897)	(0.0186)	(0.1815)	(0.1147)	(0.1890)
Lev	0.1168	0.0846	0.0673	−0.0082	−0.8576**	0.1361
	(0.6183)	(0.6267)	(0.0462)	(0.6007)	(0.3290)	(0.6263)
Growth	0.0150	0.0197	0.0247***	−0.0143	−0.0100	0.0203
	(0.0184)	(0.0193)	(0.0064)	(0.0255)	(0.0202)	(0.0195)
Board	−0.2468	−0.2241	0.0103	−0.2383	−0.3002	−0.2061
	(0.1963)	(0.1936)	(0.0193)	(0.1915)	(0.2981)	(0.1931)
Dual	0.0043	0.0038	−0.0012	0.0054	0.0440	0.0012
	(0.0551)	(0.0540)	(0.0071)	(0.0524)	(0.0947)	(0.0542)
Sh	−1.2538*	−1.3991**	−0.0309	−1.3565**	0.2393	−1.4134**
	(0.6879)	(0.6773)	(0.0916)	(0.6526)	(0.9157)	(0.6657)
Soe	−0.1185	−0.0806	−0.0120	−0.0641	−0.0756	−0.0761
	(0.2404)	(0.2490)	(0.0168)	(0.2314)	(0.0976)	(0.2487)
Age	0.4358	0.3647	0.0228	0.3333	−0.2867	0.3819
	(0.4625)	(0.4199)	(0.0450)	(0.3916)	(1.6265)	(0.4065)
_cons	19.9084***	19.8851***	0.6691*	18.9635***	2.2843	19.7478***
	(4.2659)	(4.2546)	(0.3942)	(4.1054)	(5.4666)	(4.2322)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
N	1114	1114	1114	1114	1114	1114
R^2	0.3075	0.3196	0.1317	0.3349	0.1965	0.3249

Variable definitions are in Table 1. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

capabilities. The application of AI technology in the financial industry has broad prospects, and it is of great significance to the optimization and innovation of financial services. Financial institutions should actively embrace AI technology, improve service levels and market competitiveness, and continuously increase their economic value of AI technology.

4.3. Analysis of empirical results: moderating effects

Models 1–3 in Table 4 show the moderating effect of regulatory pressure on the relationship between AI technology and the digital operational capability of enterprises. Model 1 shows that AI technology (*AI*) was significantly positively correlated with digital operational capability (*DOC*) ($\beta = 0.0088$, $p < 0.05$), but the impact of regulatory pressure (*RP*) was not significant. According to Model 2, the impact of the cross-multiplier term of AI technology and regulatory pressure (*AIRP*) was not significant, as was the effect of the cross-multiplier term of centrality (*AIRP_c*) (Model 3). This proves that regulatory pressure from external financial regulatory authorities does not enhance the facilitating effect of AI technology on digital operational capability; thus, Hypothesis 3a cannot be verified.

Models 4–6 in Table 4 show the moderating effect of imitation

Table 4
Moderating effects.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	DOC	DOC	DOC	SDC	SDC	SDC
AI	0.0088**	0.0087**	0.0088**	0.1630***	0.2740***	0.1515***
	(0.0038)	(0.0042)	(0.0039)	(0.0482)	(0.0794)	(0.0486)
RP	−0.1740	−0.1775	−0.1731			
	(0.3568)	(0.5277)	(0.3225)			
IP				0.4209	0.2023	0.8989*
				(0.4116)	(0.3471)	(0.4853)
AIRP		0.0043				
		(0.2799)				
AIRP_c			0.0043			
			(0.2799)			
AIIP					0.6831*	
					(0.3771)	
AIIP_c						0.6831*
						(0.3771)
_cons	0.6760*	0.6761*	0.6761*	2.6714	2.3765	2.5013
	(0.3976)	(0.3983)	(0.3982)	(5.4234)	(5.3132)	(5.3256)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
N	1114	1114	1114	1114	1114	1114
R^2	0.1312	0.1304	0.1304	0.1965	0.2026	0.2026

Variable definitions are in Table 1. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

pressure on the relationship between AI technology and enterprise service derivative capability. Model 4 shows that AI technology (*AI*) is significantly positively correlated with service derivative capability (*SDC*) ($\beta = 0.1630, p < 0.01$), but the effect of imitation pressure (*IP*) is not significant. We add the cross-multiplier term for AI technology and imitation pressure (*AIIP*) to Model 5. The results show that the regression coefficient of the cross-multiplier term of AI technology and imitation pressure (AIIP) on service derivative capability is significantly positive ($\beta = 0.6831, p < 0.1$). The effect of the cross-multiplier term of centrality (*AIIP_c*) (Model 6) is also significantly positive ($\beta = 0.6831, p < 0.1$), indicating that under the effect of imitation pressure, financial enterprises will actively exert their autonomy to carry out business innovation and further release the value of technology, validating Hypothesis 3b.

This illustrates that financial firms can achieve better performance using AI in a highly competitive industrial environment. In addition, because of the time required to adjust the regulatory system and policies, the speed of introduction of government policies and regulatory measures at this stage lags behind that of stimulating innovation through market competition to a certain extent, resulting in the failure of financial supervision to show a significant immediate regulatory effect. Moreover, compared with imitative pressure, the lack of regulatory pressure is also reflected in insufficient adaptation to emerging technologies or models. While new technologies bring new business models and financial service methods, they also bring new risks and challenges, requiring regulators to constantly update the regulatory means and methods to deal with them, which is a long-term process of continuous exploration and trial and error. Therefore, at present, China is open to the development of artificial intelligence technology and relies more on market competition than financial supervision to stimulate the development momentum of the industry and carry out in-depth exploration of technology applications.

5. Robustness testing

5.1. 2SLS estimation

Considering that there is still an endogeneity problem between the level of AI technology and the financial performance of enterprises, the better the performance of enterprises, the more willing they are to use technical tools to carry out internal business governance to capture the value demands of more stakeholders and, thus, more likely to introduce AI technology for intelligent transformation. Therefore, this study further alleviates the endogeneity problem using two-stage estimation. We use the lag term of endogenous variables as a commonly used approach. We consider that as the development of artificial intelligence has a time trend; the current level of artificial intelligence technology has a strong correlation with its future development.

2SLS estimation method is then used to retest the regression model. The results (Table 5) show that the lagged variables are significantly positively correlated with AI at the 1 % level in the first stage. Moreover, the F-statistic is greater than 10, and so, while not strictly an instrumental variable, it passes the weak 'instrumental variable' test. The table shows that in the second phase, AI technology's (AI) impact on financial performance (CFP) is still positive and significant at the 1 % level, indicating that the key results do not change after controlling for endogenous problems.

5.2. Heckman testing

We use Heckman's two-stage regression to alleviate endogeneity problems from selection bias. In the first stage of the regression, the inverse Mills ratio (*IMR*) is obtained by considering whether the company disclosed AI-related information (*AI_dummy*) as the explanatory variable, taking all the control variables of the main regression as explanatory variables, and using the probit model to perform the regression. In the second regression stage, the IMR estimated in the first stage is added to the regression model. The results (Table 5) show that the regression coefficient of AI technology (*AI*) is still significantly positive at 1 % ($\beta = 0.1010, p < 0.01$), consistent with the benchmark regression results.

Table 5
Endogeneity testing.

Variables	First Stage	Second Stage	Heckman
	AI	CFP	CFP
L.AI	0.265***		
	(7.7717)		
AI		0.408***	0.1010***
		(4.1281)	(0.0226)
IMR			0.2237***
			(0.0658)
Size	−0.130	−0.589***	−0.6234***
	(−1.3073)	(−7.5651)	(0.0584)
Lev	0.194	−0.567**	0.4335
	(0.5307)	(−2.0200)	(0.3017)
Growth	−0.044	0.070**	−0.0076
	(−1.0709)	(2.2060)	(0.0285)
Board	−0.236	−0.072	−0.3589**
	(−1.0734)	(−0.4206)	(0.1442)
Dual	−0.011	−0.032	0.0708
	(−0.1472)	(−0.5336)	(0.0552)
Sh	0.496	−1.414***	−1.6817***
	(0.7739)	(−2.8268)	(0.3749)
SOE	−0.274	−0.013	0.0282
	(−1.4045)	(−0.0865)	(0.1386)
Age	0.275	0.320	0.8760**
	(0.4551)	(0.6861)	(0.3976)
_cons			16.5926***
			(2.1398)
Individual	Yes	Yes	Yes
Year	Yes	Yes	Yes
N	941	941	877
R^2		0.210	0.4224

Variable definitions are in Table 1. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

5.3. Alternative independent variables

To further verify the robustness of the regression results, we replace some explanatory variables with alternatives. Based on the Management Discussion and Analysis (*MD&A*) section of annual reports, we construct a proxy indicator of corporate AI, the natural logarithm of the number of AI keywords in the *MD&A* section, used to measure the level of AI technology for the test results. Table 6 shows the results of this robustness testing. The regression coefficient of AI technology on financial performance is still significantly positive ($\beta = 0.0939, p < 0.01$), and the results still support the conclusions of this paper.

5.4. Alternative dependent variables

Tobin Q evaluates an enterprise's investments and market performance. As there are alternative ways to measure the Tobin Q, we calculate the Tobin Q- value by dividing the market value by the difference between total assets and net intangible assets. We employ this measure as a substitute variable for financial performance. The results show that the regression performance of AI technology on financial performance is significantly positive ($\beta = 0.0958$, $p < 0.01$), confirming the robustness of the results.

5.5. Alternating control variables

As only micro-control variables at the enterprise level are considered in our baseline modeling, we add macro-control variables at the provincial level for robustness. We add gross domestic product (*GDP*) and

Table 6
Robustness test results.

Variables	Alternate independent variable	Alternate dependent variable	Alternate control variable	Change sample interval
	CFP	CFP2	CFP	CFP
AI2	0.0939***			
	(0.0358)			
AI		0.0958**	0.1011**	0.1167**
		(0.0424)	(0.0393)	(0.0478)
Size	−0.7458***	−0.7298***	−0.7462***	−0.8437***
	(0.1885)	(0.2023)	(0.1904)	(0.1563)
Lev	0.0815	0.5719	0.0850	−0.1047
	(0.6215)	(0.6292)	(0.6302)	(0.4610)
Growth	0.0197	0.0072	0.0195	0.0008
	(0.0192)	(0.0205)	(0.0194)	(0.0292)
Board	−0.2222	−0.2548	−0.2231	−0.1444
	(0.1946)	(0.2291)	(0.1923)	(0.2328)
Dual	0.0057	0.0158	0.0036	0.0742
	(0.0547)	(0.0579)	(0.0535)	(0.0926)
Sh	−1.3853**	−1.9895**	−1.4064**	−1.7682**
	(0.6841)	(0.9060)	(0.6692)	(0.7340)
SOE	−0.0807	−0.0471	−0.0791	−0.0597
	(0.2473)	(0.2425)	(0.2502)	(0.2821)
Age	0.3570	0.6389	0.3653	0.7898
	(0.4157)	(0.4150)	(0.4149)	(0.5252)
GDP			0.0000	
			(0.0000)	
CPI			−0.0050	
			(0.0436)	
_cons	19.8803***	18.5443***	20.3793***	21.0874***
	(4.2336)	(4.5707)	(5.2504)	(3.4081)
Individual	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
N	1114	1114	1114	716
R^2	0.3178	0.2737	0.3184	0.3728

Variables are defined in Table 1. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

consumer price index (*CPI*) to supplement the original control variables with data from *CSMAR*. The remaining control variables, indicating bank characteristics, are consistent with those described in the previous section. The regression analysis results are consistent with baseline results.

5.6. Alternative sample period

The sample period of this study is from 2011 to 2022. However, the COVID-19 pandemic of 2019–2020 significantly impacted financial enterprises. Therefore, we select data from 2011 to 2019 for additional empirical analysis. These regression results are consistent with baseline results.

6. Conclusions

The rapid development of AI technology and its wide application in the financial field has boosted the intelligent transformation of financial enterprises and upgraded their service models and customer experiences. Concomitantly, in the face of developmental imbalance in the finance industry, the development and application of AI technology will become an important way for firms to differentiate.

We conduct regression analyses of 152 Chinese-listed financial enterprises from 2011 to 2022, finding that AI technology effectively empowers financial enterprises to enhance their financial performance. Therefore, financial enterprises should consider digital technologies' economic value and practical significance. In addition, in the face of common problems in the process of enterprise digital transformation, based on the theoretical perspective of corporate governance, we further explored the transmission mechanism of business process governance from the two dimensions of quantity and quality. Specifically, AI technology can achieve governance effects, such as improving financial product quality and optimizing existing business processes, by strengthening the enterprise's digital operational capability, which enhances the enterprise's operational efficiency. It also encourages financial enterprises to seek customer mining and horizontal business expansion by enhancing their service derivative capability. These results confirm the key role of AI technology in corporate governance and performance improvement. In this regard, financial enterprises should increase AI development to achieve process optimization, model innovation, and intelligent upgrading of corporate businesses, ultimately improving performance and creating excess economic benefits.

Based on institutional theory, this study enriches the boundary conditions of the mechanism by which financial firms' AI technology levels affect business process governance. Specifically, by considering regulatory and imitation pressures, we highlight the contextual roles of financial regulation and the degree of industry competition in AI-technology-driven improvement of business process governance. Our results show that financial firms perform better when using AI in highly competitive industry environments. This further illustrates that in the early stage of technological growth, the market has become a key force in the development of the AI industry, and the Chinese government has opened up policies and encouraged the financial sector to explore boldly and act quickly. In this regard, in a highly competitive market environment, the financial industry should give full play to the advantages of large-scale markets and actively carry out business innovation; promote the dynamic role of AI in corporate governance, economic activities, and scenario applications; and continue to open up new space for the development of the AI industry. For financial regulatory authorities, it is impossible to completely relax supervision. Yet, they can adopt the development model of reform while using (under the premise of controllable risks) innovative tools such as regulatory sandboxes and trying to use AI technology to improve regulatory efficiency.

This study has limitations, as we focus on listed financial companies. Owing to sample size limitations, different sizes and types of financial enterprises are not distinguished. More in-depth research can be carried out to explore possible group differences. Further, the discussion of the internal mechanism of AI technology on service performance is based on the perspective of corporate governance, while the empowerment path of AI technology is likely more complex.

Data availability

Data will be made available on request.

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